

Sharmistha Jat

London, United Kingdom
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INTERESTS	Computational Neuroscience, Natural Language Processing and Machine Learning
EDUCATION	<i>PhD</i>, Computer Science, 2015 - July 2021 (Maternity Leave, July 2019 - June 2020) Indian Institute of Science (IISc), Bangalore, India Thesis title: Relating Representations in Deep Neural Networks and the Brain Advisor: Prof. Partha Talukdar (IISc) and Prof. Tom Mitchell (CMU) <i>MTech</i>, Computer Science, 2011-2013 Indian Institute of Technology Madras (IITM), Chennai, India Advisor: Prof. Balaraman Ravindran <i>BE</i>, Computer Engineering, 2005-2009 Maharashtra Institute of Technology COE, Pune, India
WORK EXPERIENCE	Xerox Research Centre India (XRCI), Bangalore, Oct 2013- July 2015 <i>Research Engineer, Analytics Lab</i> Member of social media analytics and transportation research groups. Worked closely with Research Scientists to formulate and solve real-world problems using machine learning techniques. Developed production-level prototypes to integrate research tools into the business and published two papers and four patents Morgan Stanley, Mumbai, Jan 2010 - July 2011 <i>Senior Associate, Technology Division</i> The role involved working with New York based Fed reporting team for developing a global reporting platform. I helped design, develop, optimize, and maintain end-to-end systems to process massive data streams. Worked actively in the frontend, middle-tier web services, and backend database management
INTERNSHIP EXPERIENCE	Carnegie Mellon University (CMU), Pittsburgh Sept 2017- April 2018, Sept 2018 - March 2019 Advisor: Prof. Tom Mitchell
PUBLICATIONS	Sharmistha Jat, Partha Talukdar, Tom Mitchell, Word Salience during Sentence Reading, Accepted at Human Brain Mapping (oHBM), Seoul, Korea, 2021 Sharmistha Jat, Partha Talukdar, Tom Mitchell, Empirical Evaluation of Composition in Deep Neural Networks using Brain Data, under review 2021 Sharmistha Jat, Erika J C Laing, Partha Talukdar, Tom Mitchell, Spatio-temporal Characteristics of Noun and Verb Processing during Sentence Comprehension in the Brain, 2020 [under review, preprint available] Sharmistha Jat, Hao Tang, Partha Talukdar, Tom Mitchell, Relating Simple Sentence Representations in Deep Neural Networks and Brain, Accepted at Association for Computational Linguistics (ACL), Florence, 2019

S. Kumar, Sharmistha Jat, Karan Saxena, Partha Talukdar, **Zero-shot Word Sense Disambiguation using Sense Definition Embeddings**, Accepted at Association for Computational Linguistics (ACL), Florence, 2019 [\[outstanding paper award\]](#)

Sharmistha Jat, Tara Pirnia, Tom Mitchell, **Dynamic Time Warping for Brain MEG Analysis**, Accepted at Human Brain Mapping (oHBM), Rome, 2019 [\[travel stipend award\]](#)

Nicole Rafidi, Mariya Toneva, Dan Schwartz, Sharmistha Jat, Tom Mitchell, **MEG Representational Similarity Analysis Implicates Hierarchical Integration in Sentence Processing**, Organization for Human Brain Mapping, OHBM 2018, Singapore, June 17-21

Sharmistha Jat, Siddhesh Khandelwal and Partha Talukdar, **Improving Distantly Supervised Relation Extraction using Word and Entity Based Attention**, Automated Knowledge Base Construction Workshop, NIPS 2017, Long Beach, USA, Dec 4, 2017

Tushar Nagarajan, Sharmistha and Partha Talukdar, **CANDiS: Coupled & Attention-Driven Neural Distant Supervision**, The First Women and Underrepresented Minorities in NLP Workshop, WiNLP, ACL 2017, Vancouver, Canada, July 30, 2017

Lavanya Sita Tekumalla, Sharmistha, **IISCNLP at SemEval-2016 Task 2: Interpretable STS with ILP based Multiple Chunk Aligner**, workshop on Semantic Textual Similarity (SemEval) NAACL 2016, San Diego, 707-712

Himanshu Sharad Bhatt, Sandipan Dandapat, Peddamuthu Balaji, Shourya Roy, Sharmistha and Deepali Semwal, **SODA:Service Oriented Domain Adaptation Architecture for Microblog Categorization**, Proceedings of NAACL 2016, San Diego, 77-81

Anuj Mahajan, Sharmistha, Shourya Roy, **Feature Selection for Short Text Classification using Wavelet Packet Transform**, Proceedings of the 19th Conference on Computational Natural Language Learning, CoNLL 2015, Beijing, China, July 30-31, 2015, ISBN - [978-1-941643-77-8], 321-326

Sharmistha, Madhur Amilkanthwar, Shankar Balachandran, **Augmentation of Programs with CUDA Streams**, 10th IEEE International Symposium on Parallel and Distributed Processing with Applications, ISPA 2012, Leganes, Madrid, Spain, July 10-13, 2012, ISBN - [978-1-4673-1631-6], 855-856

PATENT GRANTED Sharmistha Jat, Anuj Mahajan, Shourya Roy, **Method and System for Predicting Requirements of a User for Resources Over a Computer Network**, 2019, United States Patent 10417578

Sharmistha Jat, Koyel Mukherjee, Narayanan U. Edakunni, Pallavi Manohar, **Method and system for recommending one or more vehicles for one or more requesters**, 2018, USPTO Patent Number 9978111

Narayanan U. Edakunni, Sharmistha Jat, **Method and system for determining effect of weather conditions on transportation networks**, 2018, USPTO Patent Number 10062282

Koyel Mukherjee, Ankit Kumar, Pallavi Manohar, Narayanan U. Edakunni, Sharmistha Jat,

Method and system for scheduling vehicles along routes in a transportation system, 2017, USPTO Patent Number 9746332

COURSES	Machine learning, Computational Methods of Optimisation, Stochastic Methods and Applications, Linear Algebra, Natural Language Understanding, Social Network Analysis, Knowledge Representation and Reasoning, Kernel Methods in Pattern Recognition
Computer Skills	Programming languages (C, C++, C#, Matlab, Java, Python), Frameworks (PyTorch, NumPy, SciPy), Databases (MySQL, MongoDB)
Links	http://sharmistha.co.in/ https://www.linkedin.com/in/sharmistha-jat-126b9316/ https://github.com/SharmisthaJat https://scholar.google.co.in/citations?user=YVZONDwAAAAJ&hl=en&oi=ao
POSITION OF RESPONSIBILITY	Reviewer for ICLR 2021, CoNLL 2019, NIPS 2018, KDD 2018, XRCI Open Innovation Conference 2015 Teaching Assistant for Machine Learning (EO-270), Indian Institute of Science (IISc), January- April 2017 Organized Girl Student Empowerment workshop at Department of CDS, IISc, in collaboration with CHETANA Girls organisation. Tutorial, “Predictive Modeling Using R and Python” at Grace Hopper Conference (GHC), India, 8th December 2016 Teaching Assistant for Social Network Analysis (CS6012), Indian Institute of Technology Madras (IIT Madras), January-April 2013 Member of Charity Committee, Morgan Stanley 2010-2011
HONORS and AWARDS	BrainHub CMU-IISc Fellowship 2017-2018, 2018-2019 AAAI 2019 travel grant, OHBM 2019 travel stipend, NAACL 2016 travel grant Selected to attend EMEA workshop for Brain Inspired Computing 2018 and Google NLP summit 2017 Top 6% in KAGGLE. Member of MALL Lab team which stood 10th among 170 teams for ALLEN-AI Kaggle challenge, demo: http://malllabiisc.github.io/demos/ Best Poster Award at Grace Hopper Celebration India 2016, GHCI is the India’s largest gathering of women technologists with almost 6000 attendees. Second position winner at Yahoo Hacku, a national level coding competition 2012 Third position winner in Code-Obfuscation coding challenge at Shaastra 2011, a national level competition organised by Indian Institute of Technology Madras (IITM)